

Total No. of Questions : 8]

SEAT No. :

PE-2565

[Total No. of Pages : 3

[6583]-94

T.E. (E&TC Engineering)
POWER DEVICES & CIRCUITS
(2019 Pattern) (Semester - VI) (304194)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8
- 2) Neat diagrams and waveforms must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of nonprogrammable calculator is allowed.
- 5) Assume Suitable data if necessary

Q1) a) Single phase full bridge inverter is operated from 120V dc supply, it has a resistive load of $R = 20 \Omega$. Find: [6]

- i) rms o/p voltages at third & fifth harmonic (V_{o3} & V_{o5})
- ii) Distortion factor (DF) of 3rd harmonic component
- iii) Total harmonic distortion (THD)

b) Give classification of inverters? Explain role of feedback diodes in inverters. [4]

c) Explain working of single phase full bridge inverter (using MOSFET / IGBT) for R load with input & output waveforms. [7]

OR

Q2) a) Draw a three phase inverter for balanced star R load? Explain its operation of 180° mode with gate signals & output waveforms. [11]

b) What are advantages of PWM techniques in inverter? Explain Multiple PWM technique with waveforms. [6]

Q3) a) A step down chopper is operated from dc supply voltage of 200V. It has resistive load with $R=20 \Omega$. When chopper operates, voltage drop across chopper is 2V. If duty cycle is 50%, calculate: [6]

- i) Average & rms o/p voltages
- ii) Average & rms o/p currents
- iii) Chopper efficiency

b) Explain with diagrams various control techniques in DC chopper operation. [6]

c) Explain with block schematic working of SMPS. [6]

P.T.O.

OR

- Q4)** a) A step up chopper is operated from 230V dc supply and it provides 500V output. If chopping frequency is 2KHz, calculate ON & Off times of chopper. [4]
- b) Explain with circuit diagram operation of step up chopper and derive an expression for its o/p voltage : $V_o = \frac{V_s}{(1-D)}$ where D is duty cycle. [8]
- c) Explain operation of step down chopper with R-L load for continuous & discontinuous current mode. [6]
- Q5)** a) What is snubber circuit? Explain with circuit diagram working of snubber circuit. [6]
- b) Why isolation is required in power electronic circuits? Explain with neat diagram working of isolation transformer. [7]
- c) For a thyristor, Maximum junction temperature is 200°C. The thermal resistances are $\theta_{JC} = 0.16^\circ\text{C/W}$, $\theta_{CS} = 0.08^\circ\text{C/W}$. For heat sink temperature of 90°C, calculate total average power loss in thyristor - sink combination.
- If heat sink temperature is reduced to 70°C, find new total average power loss in thyristor - sink combination [4]

OR

- Q6)** a) What is the need of resonant converter? Explain ZCS resonant converter with circuit & waveforms. [7]
- b) What are various cooling techniques used in power electronic circuits? Explain in brief any one with thermal equivalent. [4]
- c) What is EMI? What are various sources? Explain minimizing techniques of EMI. [6]

- Q7)** a) Explain with circuit diagram & various waveforms, how single phase full wave AC voltage controller is used as a fan regulator. [6]
- b) Explain various battery charging models for EVs [6]
- c) Explain operation of On-line UPS with block schematic. [6]

OR

- Q8)** a) Explain single phase semi converter drive for single phase separately excited dc motor. [6]
- b) Why driver is required for LED lamp? Explain with suitable circuit diagram working of a LED lamp driver. [5]
- c) Explain with neat diagram variable voltage / frequency three phase induction motor drive. [7]

